

Complete Product & Architecture Guide

Every menu, every option, the full system architecture, and the end-to-end flow of data — with a clear split of what Owners and Employees can do.

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1. What DecisionOS is

DecisionOS is an AI operating system for founder-led small & medium businesses. It captures spoken or typed decisions, reads incoming documents/invoices/messages, turns them into structured tasks and workflows, and gives the whole team one shared operational brain. It is **multi-tenant** (each company is an isolated workspace) with **strict role-based access**, and every screen adapts to the company's industry using an AI-generated **Operating Model** (workflow pipelines + task categories) and **Business Vocabulary** (the words your industry actually uses).

The one-line mental model

Anything that happens in the business enters through **Capture** or **Decision Desk** → AI structures and routes it → it becomes **Tasks / Decisions / Workflows / Finance** records → everyone acts in **My Work** → the **Company Brain** remembers it → the **CEO Brief** and **Notifications** surface what needs attention.

2. System architecture

DecisionOS is a three-tier web application with an AI layer on top.

Layer	Technology	Responsibility
Frontend	React, TailwindCSS, Shadcn UI, React Router, TanStack Query, react-i18next	Renders all menus/pages; per-user language (EN/HI/TA); talks to the API via a single axios client using REACT_APP_BACKEND_URL.
Backend	FastAPI (Python), all routes prefixed /api	Auth & RBAC, business logic, AI orchestration, WhatsApp webhook, follow-up/escalation engine, file storage.

Layer	Technology	Responsibility
Database	MongoDB (Motor async)	Stores every entity keyed by <code>tenant_id</code> for multi-tenant isolation. IDs are UUID strings.
AI layer	Anthropic Claude, Google Gemini, OpenAI (transcribe/vision) with Emergent Universal LLM key as fallback (see <code>core.py</code>)	Transcription, classification, extraction, structuring, finance analysis, Ask/Search answers, operating-model & lexicon generation.

Key architectural ideas

- **Multi-tenancy:** every document carries a `tenant_id`. Users only ever read/write their own tenant's data.
- **Dynamic UI:** tabs, pipeline stages and task categories are read from `tenant.operating_model`; contact/vendor labels from `tenant.lexicon` — so a textile mill and a clinic see different terminology on the same code.
- **Auth:** JWT bearer token (also set as a cookie) issued on register/login; passwords hashed with `bcrypt`.
- **Hot paths run in the background:** long AI jobs (transcription, structuring) are processed asynchronously; the UI shows a 'structuring...' banner and updates when ready.
- **Single source of config:** URLs, DB and keys come only from environment variables (frontend `.env` → `REACT_APP_BACKEND_URL`; backend `.env` → `MONGO_URL/DB_NAME`).

3. Roles & access model

There are two conceptual levels: the **Owner** (full control) and **Employees** (granular, module-level access). Access is enforced on **both** the frontend (menus hide) and the backend (endpoints reject unauthorised calls). Employees are given a **role** (e.g. Sales, Finance, Operations — roles are customisable per company) plus an explicit list of **module permissions**.

3.1 The 14 module permissions

Every menu/action maps to one of these permission keys. Owners implicitly hold all of them.

Permission key	Grants access to
inbox	Decision Desk (unified inbox)
voice_capture	The voice/text capture box on the Decision Desk
data_input	Capture menu (uploads, CSV/doc ingest, WhatsApp intake)
people	People menu (team + customers + vendors)
finance	Finance 360° (invoices, payments)
ledger	Finance Ledger (expenses, assets, inventory)
workflows	Workflow pipelines
tasks	Tasks
brain	Company Brain (search)
ask	Ask AI
approvals	Approve tasks
decisions_approve	Approve decisions
leave_approve	Approve leave requests
team_manage	Manage team, roles, and company operating settings

Default permissions per role (owner can override any of these per employee):

- **Base (all employees):** inbox, data_input, people, workflows, tasks, brain, ask.
- **Sales:** the base set.
- **Finance:** base set + finance + ledger.
- **Owner:** all 14 permissions, always.

3.2 Owner vs Employee — side by side

Capability	Owner	Employee
See all menus	Yes — nothing is hidden	Only menus their permissions allow
Settings menu	Yes (owner-only)	No
Manage team & roles / grant permissions	Yes	Only if given team_manage
Promote someone to co-owner	Yes (≥1 owner must remain)	No
Approve decisions / tasks / leave	Yes	Only with matching approve permission
CEO Brief fires/counters & Operating Score	Yes (owner-focused)	Limited / hidden owner-only pages

Capability	Owner	Employee
Edit Operating Model, Vocabulary, Approval rules	Yes	No
See finance	Yes	Only with finance/ledger permission
Own tasks, workflows, brain, capture	Yes	Per permission (base set by default)

How hiding works in practice

The sidebar builds itself from a permission-filtered list. If an employee lacks people, the People menu simply isn't rendered, its route is protected, and any direct API call is rejected with 403. Owner-only pages (Settings, Journal, Operating Score) are gated by ownerOnly.

4. Onboarding — how an Owner's workspace is created

Signing up creates the tenant (company) and the first Owner in one step:

- Owner registers with **company name, their name, email, password, industry** and a short **business description**.
- AI immediately generates a tailored **Operating Model** (workflow pipelines with stages + task categories), a **Business Vocabulary** (lexicon), suggested **team roles/departments**, example **products/services**, **operational task templates** and **approval rules** — all specific to the industry.
- A workspace summary is returned (departments, workflows, operational tasks, approval rules counts).
- The Owner then invites employees, assigns each a role + permissions, and can edit any generated template in **Settings**.
- Employees join via an **invite link** (email + password) or as **mobile-OTP (passwordless)** users.

5. End-to-end flow of data

This is the heart of DecisionOS. Follow one item from entry to memory:

A. Entry (many front doors)

- **Decision Desk:** Owner/employee taps the mic and speaks, or types a directive.
- **Capture menu:** upload a document/image/PDF, import a CSV, or paste text.
- **WhatsApp:** a message/photo sent to the workspace number hits the WhatsApp webhook.
- **Meeting Notes:** a recorded/pasted meeting.
- **Voice notes / complaints / contacts** can also originate items.

B. AI extraction pipeline

- Audio is **transcribed** (OpenAI transcribe). Images/PDFs are **read** (vision).
- AI **classifies** the item (decision, task, invoice, payment, complaint, contact...) and **structures** it into a clean summary with fields, amounts, dates and suggested owners.
- A **confidence score** decides routing: high-confidence items can auto-process; lower-confidence items are flagged for human review.

C. Department review queue (Captures)

- A **draft capture** is created and routed to the correct reviewer: money items → the tenant's finance/accounts role (by `reviewer_perm=finance`); operational items → the relevant department.
- The Owner sees everything; an employee sees a draft if their role matches the reviewer role OR they hold the reviewer permission.
- Reviewer actions: **Approve**, **Reject**, **Clarify** (ask AI/askback), or **Reassign** to another member/role.

D. Commit — records are created

- On approval the draft becomes real records: **Decisions** (with linked **Tasks**), **Workflow** cards, **Expenses/Assets/Inventory**, **Invoices/Payments**, **Contacts**, or **Complaints**.
- High-value money items respect the company's **threshold** and **owner sign-off** rules from Settings.

E. Action — My Work

- Tasks land in the assignee's **My Work**. AI can **prioritise** them. Workflow cards advance stage-by-stage.
- Assignees respond, add updates, attach files, or generate an AI **execution plan**.

F. Follow-up & escalation engine

- A background engine watches due dates and SLAs, sends **follow-ups**, and **escalates** overdue/at-risk items — producing **Notifications**.

G. Surface — CEO Brief & Notifications

- The **CEO Brief** aggregates the day into a headline + 'fires to put out', with deep links to each record.
- The **bell** shows follow-ups, escalations and mentions; clicking one jumps to the item.

H. Memory — Company Brain

- Every decision, task, workflow, contact and 'company memory' note is indexed so **Ask** gives grounded answers with citations and **Search** finds exact records.

Leave & absence — a parallel flow

When someone requests leave or reports an absence, AI computes the **impact** (which tasks/workflows are at risk and who should cover), routes it to the configured **leave approvers**, and can apply reassignments in one click.

6. Menu-by-menu deep dive (every option)

Below is each sidebar menu with all of its sub-tabs, buttons and controls. Interactive elements are named exactly as they appear/behave in the product.

Top bar & global tools

Route: (all screens) Access: Everyone

Always visible controls at the top and around the app.

Controls

- **Send Daily Digest** — emails the current CEO Brief to you/the team.
 - **Language switcher** (globe) — English / ■■■■■■ / ■■■■■■; applies to you only.
 - **Light / Dark theme** toggle.
 - **Notifications bell** — unread count badge; dropdown of the latest items; click to open the item; 'Mark all read'.
 - **Profile** — open profile dialog; **Sign out**.
 - **Voice mic buttons** — dictate instead of typing in most inputs.
 - **Mobile bottom tab bar** — quick access to Desk, Brief, Work, People, Brain.
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6.1 Decision Desk

Route: /inbox Access: Requires 'inbox' permission

The unified inbox — the single front door to capture and triage everything.

Capture box

- **Voice record** button with a live **timer**; controls to **pause**, **resume**, **cancel** and **finalise** the recording.
- **Type a directive** text box + **submit** (e.g. 'Tell sales to send the revised quote to the Delhi retailer by Friday...').
- A **language selector** for the capture, and a **structuring banner / thinking overlay** while AI works — you can keep going.

Triage sections

- **Decision Approvals** — approve / reject / discuss AI-extracted decisions.
 - **Needs Your Attention** — items AI flagged for you.
 - **Tasks & Activity feed** — live stream with **filters**; on mobile **swipe right = view**, **swipe left = dismiss** (a swipe hint is shown).
 - **Clarify panel** — when AI needs more info: submit answer, skip, or cancel.
 - **Owner-only notice** appears where an action is limited to owners.
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6.2 CEO Brief

Route: /brief Access: All users (owner-focused; some counters owner-only)

A daily AI executive summary of what needs attention.

On the page

- **Brief controls** — switch period (e.g. morning/evening).
 - **'Fires to put out today'** with a **fires count** and clickable **fire rows** (deep-link to the exact record).
 - **Open Work Coach** — AI coaching on priorities.
 - **Open Journal** — the owner's decision journal (owner-only).
 - **Operating Score** — a health score for how the business is running (owner-only).
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6.3 My Work

Route: /my-work Access: All users (items filtered to what you can access)

Your personal operational cockpit: tasks, board, workflows and leave.

Scope & priority

- **My Tasks** vs **All Tasks** scope toggle.
- **AI Priority** toggle — AI scores & orders tasks; each shows a **priority score** with bars.

View toggle (4 views)

- **My Work** — task list grouped by category.
- **Board** — kanban by status.
- **Workflows** — pipeline cards (see 6.x); advance stage, create, delete.
- **Leave** — opens the Leave & Absence view.

Per-task actions

- Respond, add updates, attach files, mark done, reassign, approve/reject (with permission), generate/edit an AI **execution plan** and ask about steps.
 - An **access-restricted banner** shows if you open an item you can't access.
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6.4 People

Route: /contacts Access: Requires 'people' permission

CRM-lite for your team, customers and vendors. Tab labels adapt to your industry.

Tabs

- **Employees** (needs team_manage) — add users; set role, permissions, reporting manager; invite links.
- **Customers** — customer/dealer records.
- **Vendors** — supplier/vendor records.

Contact tools

- **Add contact**, **search**, filter by **status** and **type**.
- Contact form: **name**, **company**, **email**, **phone**, **birthday**, **status**, **assigned owner**, **type**.
- Log a **complaint** against a contact (text + save).

Contact profile page

- Header with back button, a **relationship card** and **relationship signals**, and a **Re-score** button (AI re-evaluates the relationship health).
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6.5 Company Brain

Route: /brain Access: Requires 'brain' permission (Ask uses 'ask')

The shared memory of the business.

Ask tab

- Ask a question in plain language; AI answers grounded in your data with **source citations**.
- Suggestion chips (e.g. 'What purchases need my approval?'), a text box, **voice dictation**, and Send.

Search tab

- Full-text search across **Decisions**, **Tasks**, **Workflows**, **Contacts** and **Company Memory**; results grouped by type with counts.
 - Voice dictation supported. Rule of thumb: **Ask = answer**, **Search = records**.
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6.6 Finance

Route: /ledger Access: Requires 'ledger' or 'finance' permission

All money in one place with an AI finance analyst.

Tabs

- **Overview** — KPIs (Total Spend, Outstanding, Asset Value, Inventory Value), Monthly Spend chart, By-Category donut, Top Vendors bars.
- **Expenses, Assets, Inventory** — tables with delete; add via dialog or by attaching a bill/PDF that AI reads.

AI Finance Brief / Analysis (each tab)

- Headline + prioritised **action items** tagged Urgent / Important / FYI (expandable).
- From an insight: **Create task** (assign to member/role, set priority) or **Ask AI** about it.
- **Refresh** re-runs the analysis; **Ask AI** box for finance questions.

Add dialogs

- **Expense**: attach bill, title, amount, status (paid/unpaid), vendor, AI **suggest category**, date, notes.
- **Asset**: name, purchase amount, category, vendor, purchase date, status (active/maintenance/disposed).
- **Inventory**: item, SKU, quantity, unit, unit cost, category, vendor (value = qty × unit cost).
- Approved purchase bills & outgoing payments auto-flow in from WhatsApp/uploads.

6.7 Capture

Route: /ingest Access: Requires 'data_input' permission

The intake tool that feeds the AI pipeline.

Import tab

- **Upload document** (image/PDF) drop-zone + file picker.
- **Import CSV** drop-zone + file picker.
- A **direction banner** and an **own-company warning** guard against mis-ingesting your own data.
- **WhatsApp card**: connection status, **QR link**, token-error hint, and a **logs list** with refresh.

Review tab

- A **badge** shows pending items. Sub-tabs to review extracted **Invoices, Payments, Contacts, Tasks** before committing.
- **Records** tables for Invoices and Payments; commit an ingestion to finalise.

6.8 Meeting Notes

Route: /meetings Access: All users

Turn meetings into minutes and actions.

Options

- **Record** a meeting by voice (recorder UI), or toggle **paste/type** to submit a transcript.
- AI writes **Key Points** and extracts decisions/tasks; everything becomes searchable in Company Brain.
- A **meetings list** of past sessions; open any to view its minutes.

6.9 Settings

Route: /settings Access: Owner only

Where the Owner configures how the whole company operates.

Cards

- **Company Details** — name, industry, description (sharpens AI), currency, region, GST, branches.
- **Business Vocabulary** — customer/vendor labels, task-type/department labels; **regenerate** with AI.
- **Operating Model** — edit **workflow pipelines** (add/rename/delete pipelines & stages, set the owner sign-off stage) and **task categories**; add roles; **regenerate** with AI.
- **Team Roles** — create/rename/delete roles; **Products & Services** list.
- **Approval rules** — company rules that govern sign-off.

- **Language** — your interface language (per-user).
 - **Your Profile** — name, mobile, sign-in email.
 - **Password & Security** — change your password (current → new → confirm).
 - **Money & Approvals** — default currency, high-value threshold, and 'require owner sign-off above threshold' toggle.
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6.10 Additional / linked pages

Route: /journal, /operating-score, /coach, /calendar, /notifications Access: Mixed
Reachable from the CEO Brief, bell, or deep links.

Pages

- **Journal** (owner-only) — the founder's decision journal.
 - **Operating Score** (owner-only) — overall business-health score.
 - **Work Coach** — AI coaching; refreshable.
 - **Calendar** — time-based view of tasks/leave/events.
 - **Notifications** — full list with **mark-all-read**.
 - **Leave & Absence** (via My Work) — tabs: **My leaves**, **Approvals**, **Approver config**; request leave (type, from/to, half-day portion, reason), report absence (reason, note), view AI **impact** and apply-all reassignment, save approvers.
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7. Owner journey vs Employee journey

7.1 Owner — a typical day

- Open **CEO Brief** → read the headline and clear the 'fires' (each links to the record).
- Check **Decision Desk** → approve/reject AI-extracted decisions; reassign anything mis-routed.
- Review money in the **Capture** → **Review** queue and **Finance** (respecting the high-value threshold).
- Dictate new directives by voice; AI turns them into assigned tasks.
- Manage the team in **People** → **Employees**; tune rules in **Settings**.
- Ask the **Company Brain** anything; send the **Daily Digest**.

7.2 Employee — a typical day

- Lands on **Decision Desk** (or My Work if no inbox access) — sees only permitted menus.
- Works through **My Work** tasks (optionally AI-prioritised); advances **Workflow** cards.
- Captures field updates by voice/upload in **Capture**; logs customer issues in **People**.
- Requests **Leave** — AI shows who/what is impacted and routes it to approvers.
- Uses **Company Brain** to find past decisions instead of asking the founder.
- Never sees Settings, other departments' finance, or owner-only pages unless granted.

8. Core data entities

Entity	Purpose / key fields
tenant	The company workspace: name, industry, description, currency, roles, products, lexicon, operating_model, approval_rules, threshold & sign-off settings.
user	Member: name, email, phone, role, permissions[], reporting_manager_id, language, passwordless flag, password_hash.
capture	AI draft awaiting review: type, extracted fields, confidence, reviewer_role, reviewer_perm, status.
decision	Structured decision: title, summary, status, linked tasks, timeline/comments.
task	Work item: title, description, priority, status, assignee_id/role, updates, attachments, execution plan.
workflow	Pipeline card: type, title, counterparty, amount, stage, stages[], history.
expense / asset / inventory	Finance ledger records with source, attachment, amounts.
invoice / payment	Money-360 records from ingest/WhatsApp.
contact	Customer/vendor/dealer: name, company, phone, type, status, relationship score.
leave / attendance	Time-off requests & absences with AI impact.
meeting / voice-note	Transcribed sessions and their extracted actions.
memory	Company Brain notes/tags.
notification / activity	Alerts and the audit log of who did what.

9. API reference (grouped)

All endpoints are prefixed with `/api` and require a bearer token except `auth/login`, `auth/register` and the public WhatsApp webhook.

Auth & profile

POST `/auth/register`
POST `/auth/login`
POST `/auth/logout`
GET `/auth/me`
PATCH `/auth/profile`
POST `/auth/change-password`
POST `/auth/otp/request`
POST `/auth/otp/verify`
GET `/auth/invite/{token}`
POST `/auth/invite/{token}/start`

Decision Desk / Inbox

GET `/inbox`
POST `/inbox/{item_id}/status`
GET `/decisions`
GET `/decisions/{id}`
GET `/decisions/{id}/timeline`
POST `/decisions/{id}/approve`
POST `/decisions/{id}/reject`
POST `/decisions/{id}/comment`
POST `/decisions/{id}/tasks`

Captures (review queue) & ingest

GET `/captures`
GET `/captures/pending-count`
PATCH `/captures/{id}`
POST `/captures/{id}/approve`
POST `/captures/{id}/reject`
POST `/captures/{id}/clarify`
POST `/captures/{id}/reassign`
POST `/capture/clarify`
GET `/ingest`
GET `/ingest/{id}`
POST `/ingest/document`
POST `/ingest/csv`
POST `/ingest/{id}/commit`

Tasks

GET `/tasks`
GET `/tasks/{id}`
POST `/tasks`
PATCH `/tasks/{id}`
DELETE `/tasks/{id}`
POST `/tasks/prioritize`
POST `/tasks/{id}/approve`
POST `/tasks/{id}/reject`
POST `/tasks/{id}/reassign`
POST `/tasks/{id}/respond`
POST `/tasks/{id}/updates`
POST `/tasks/{id}/attachment`
POST `/tasks/{id}/clarify`
POST `/tasks/{id}/execution-plan/generate`
PATCH|DELETE `/tasks/{id}/execution-plan`
POST `/tasks/{id}/steps/ask`

Workflows

GET `/workflows`
POST `/workflows`
PATCH `/workflows/{id}/advance`

DELETE /workflows/{id}

Finance (ledger & 360)

GET /ledger/summary
GET|POST|DELETE /expenses...
expenses/with-file, assets/with-file, inventory/with-file
POST /expenses/suggest-category
GET /ledger/ai/{scope}
POST /ledger/ai/{scope}/refresh
POST /ledger/ask
GET /invoices
GET /payments

People, Company Brain, meetings, leave

GET|POST|PATCH|DELETE /contacts...
GET /contacts/{id}/profile
POST /contacts/{id}/rescore
POST /complaints
PATCH /complaints/{id}/resolve
GET /brain/search
POST /ask
GET|POST /memory
GET|POST /meetings
POST /meetings/text
POST /transcribe
POST /voice-notes
GET|POST /leaves...
POST /leaves/absence
POST /leaves/{id}/approve|reject|request-info
GET /leaves/{id}/impact

Owner config, team & system

PATCH /tenant
PATCH /tenant/settings
PATCH /tenant/lexicon
POST /tenant/lexicon/regenerate
PATCH /tenant/operating-model
POST /tenant/operating-model/regenerate
GET|POST|PATCH|DELETE /tenant/roles...
PATCH /tenant/leave-approvers
GET /users
POST /users
PATCH /users/{id}
POST /users/{id}/invite
GET|POST /invites
GET /brief
GET /brief/details
POST /brief/send-digest
GET /dashboard
GET /operating-score
GET /work-coach
POST /work-coach/refresh
GET /notifications
POST /notifications/{id}/read
POST /notifications/read-all
POST /follow-up/run
GET|POST /webhooks/whatsapp
GET /whatsapp/status
GET /whatsapp/logs

10. Integrations, AI models & security

AI & integrations

- **Anthropic Claude, Google Gemini, and OpenAI** (transcription + vision) power the AI. The **Emergent Universal LLM key** is the automatic fallback so AI keeps working even without your own keys.
- **WhatsApp Cloud API** — inbound messages/photos become captures automatically via the webhook.
- Company keys (OpenAI / Anthropic / Gemini) can be configured; otherwise the universal key is used.

Security & isolation

- **Passwords** hashed with bcrypt; never stored or returned in plain text.
- **JWT bearer** tokens for every authenticated request; passwordless members sign in via mobile OTP.
- **Tenant isolation** — every query is scoped to the caller's tenant_id.
- **Server-side permission checks** — the backend re-validates permissions on every protected route (not just the hidden menu).
- **Audit trail** — an activity log records key actions (who did what, when).

DecisionOS — one shared operational brain for founder-led teams. Multi-tenant, role-based, AI-native. This guide reflects the current product build.